LECTURE 5

Claim $\begin{bmatrix} 1 \\ -1 \end{bmatrix} \notin \operatorname{span}\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \end{bmatrix} \right\}$

Proof Suppose it were. Then $\exists\, a_1, a_2 \in \mathbb{R}$ such that

$$\begin{bmatrix} 1 \\ -1 \end{bmatrix} = a_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + a_2 \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

$$= \begin{bmatrix} a_1 + 3a_2 \\ 2a_1 + 6a_2 \end{bmatrix}$$

$$\Rightarrow \begin{aligned} 1 &= a_1 + 3a_2 \\ -1 &= 2a_1 + 6a_2 \end{aligned} \quad \overset{\times 2}{\Rightarrow} \quad \begin{aligned} 2 &= 2a_1 + 6a_2 \\ -1 &= 2a_1 + 6a_2 \end{aligned} \quad \Rightarrow \quad 2 \neq -1$$

Contradiction!

Claim $\mathbb{R}^2 = \operatorname{span}\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 4 \end{bmatrix} \right\}$

Proof We need to show $\subseteq$ and $\supseteq$.

$\subseteq$: Let $\begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2$. We need to find $a_1, a_2 \in \mathbb{R}$ such that

$$\begin{bmatrix} x \\ y \end{bmatrix} = a_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + a_2 \begin{bmatrix} -1 \\ 4 \end{bmatrix}$$

$$= \begin{bmatrix} a_1 - a_2 \\ 2a_1 - 4a_2 \end{bmatrix}$$

We want to solve $\begin{aligned} x &= a_1 - a_2 \\ y &= 2a_1 - 4a_2 \end{aligned}$ for a_1 & a_2.

Subtract 2 ① from ② $\Rightarrow y - 2x = 6a_2$.

Hence $a_2 = \dfrac{y - 2x}{6}$, $a_1 = x + a_2$

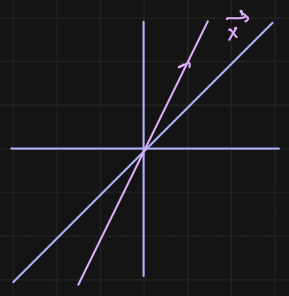
$$= x + \frac{y - 2x}{6}$$

Thus $a_1 = \dfrac{4x + y}{6}$

CONTINUED

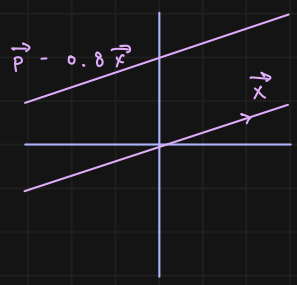
Idea

If $\vec{x} \in \mathbb{R}^n$ and $\vec{x} \neq \vec{0}$, then span $\{\vec{x}\}$ is a line through the origin.



$= \{ t\vec{x} : t \in \mathbb{R} \}$

For a line not through $\vec{0}$, we can translate.



$\vec{p} + 1.7\vec{x}$

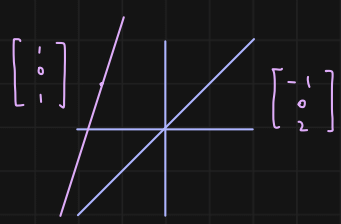
Def'n

A <u>line</u> through $\vec{p} \in \mathbb{R}^n$ with direction vector $\vec{x} \neq 0$ is given by the set $\{ \vec{p} + t\vec{x} : t \in \mathbb{R} \}$

Ex

In $\mathbb{R}^3$, the line $\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} : t \in \mathbb{R} \right\}$

passes through $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ and has direction parallel to $\begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$



Idea

Given $\vec{x}, \vec{y} \in \mathbb{R}^n$ that are not scalar multiples of each other, span $\{\vec{x}, \vec{y}\}$ is a plane through $\vec{0}$.

EX: If $\vec{x} = c\vec{y}$, then span $\{\vec{x}, \vec{y}\}$ = span $\{\vec{x}\}$

line

CONTINUED

Idea For planes not through $\vec{0}$, we can translate.

Def'n A <u>plane</u> containing the point $\vec{p} \in \mathbb{R}^n$ with direction vector $\vec{x}$ & $\vec{y}$ is given by

$$\{ \vec{p} + t\vec{x} + s\vec{y} : t, s \in \mathbb{R} \}$$

Ex In $\mathbb{R}^3$: xy - plane $\left\{ \begin{bmatrix} x \\ y \\ 0 \end{bmatrix} : x, y \in \mathbb{R} \right\}$

$$= \left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} : t, s \in \mathbb{R} \right\}$$

plane